

UNIVERSITY OF GHANA — DEPARTMENT OF COMPUTER SCIENCE
DCIT 208 — SOFTWARE ENGINEERING
D2: Software Requirements Specification

Team Name	Snart
Project Title	InspectoFleet
Client Name	Phanuel K. A. Wagba
Repository URL	https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-seg26-77
AI Assistance Disclosure	Claude (Anthropic) was used to structure this SRS, organize sections per the deliverable instructions, and draft use case and user story scaffolding. All client requirements, acceptance criteria, and validation decisions were supplied and reviewed by the team. See AI_USAGE.md for the full log.

A. Requirements Overview

A1. Purpose and Scope

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for InspectoFleet, a digital vehicle inspection and fleet-status system being built for Kal Car Rentals this semester. The document establishes the requirements baseline from which the product backlog, architecture, and implementation will be derived.

InspectoFleet replaces Kal Car Rentals' paper-based vehicle inspection process with a mobile-friendly digital workflow that captures time stamped photographic evidence of vehicle condition at handover and return. It also provides a fleet status dashboard giving fleet managers visibility into vehicle availability and inspection history.

A2. System Boundary

In scope for this semester: digital inspection workflow (mobile interface), time stamped photo evidence capture, inspection record storage and retrieval, fleet status dashboard, user authentication and role based access control.

Explicitly out of scope: physical GPS/telemetry hardware integration, payment processing, customer-facing portal, native mobile app (iOS/Android), and SMS/email notifications. The system will be designed so GPS/telemetry data can be integrated in a future phase without redesigning the core.

A3. Stakeholder Map and User Roles

- Counter Agent — performs vehicle handover/return inspections; primary user of the mobile inspection interface

- Fleet Manager — monitors vehicle status and inspection history; primary user of the fleet dashboard
- Management / Owner (Phanuel K. A. Wagba) — reviews summary reports and dispute evidence; secondary system user
- Customers — indirect stakeholders; not direct system users but directly affected by evidence captured at handover/return

A4. High-Level Narrative

When a customer collects a vehicle, a counter agent opens InspectoFleet on their mobile device, starts a new inspection linked to the rental contract and vehicle, and photographs the vehicle from required angles. Each photo is automatically time-stamped and attached to the inspection record. When the vehicle returns, the same process is repeated. If a damage dispute arises, staff can retrieve the before/after inspection records as evidence.

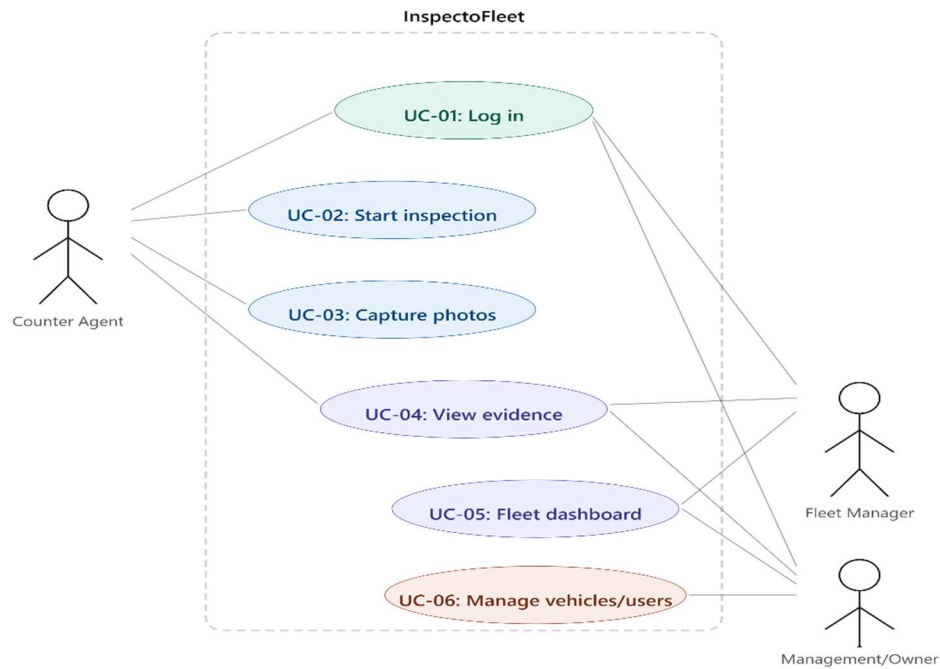
Fleet managers view a dashboard showing which vehicles are available, rented, or currently under inspection, and can browse inspection history by vehicle.

A5. Assumptions, Constraints and Dependencies

- Counter agents will use a modern smartphone browser (Chrome/Safari on Android or iOS) capable of accessing the device camera.
- Internet connectivity is available at the inspection location at least intermittently during inspection submission.
- No GPS/telemetry hardware is currently installed in Kal Car Rentals' vehicles; fleet status shown in the dashboard is derived from inspection records and manual status updates only.
- The team will decide on the final technology stack in D4; this SRS is stack-agnostic.
- Client validation of requirements is based on conversation with Phanuel K. A. Wagba (Director) and will be confirmed further before D3.

B. Use Case Model

B1. Use Case Diagram



Actors: Counter Agent, Fleet Manager, Management/Owner (secondary). System boundary: InspectoFleet.

B2. Use Case Descriptions

UC-01: Log In

Field	Detail
Use Case ID	UC-01
Name	Log In
Actors	Counter Agent, Fleet Manager, Management/Owner
Preconditions	User has an active account created by an administrator
Main Success Scenario	1. User opens InspectoFleet on their device 2. User enters email and password 3. System validates credentials 4. System redirects user to their role-appropriate home screen
Alternatives / Exceptions	3a. Invalid credentials — system displays error, user retries 3b. Account locked after 5 failed attempts — admin must reset
Postconditions	User is authenticated and has access to features appropriate to their role

UC-02: Start Vehicle Inspection

Field	Detail
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Use Case ID	UC-02
Name	Start Vehicle Inspection
Actors	Counter Agent
Preconditions	Counter Agent is logged in; vehicle exists in the system; rental contract reference is available
Main Success Scenario	1. Counter Agent selects 'New Inspection' 2. Agent selects or searches for the vehicle by registration number 3. Agent selects inspection type (Handover or Return) 4. Agent optionally links a rental contract reference 5. System creates a new inspection record and presents the photo capture screen
Alternatives / Exceptions	2a. Vehicle not found — agent can add vehicle registration manually 4a. No contract reference available — agent proceeds without it (flagged for follow-up)
Postconditions	A new inspection record exists in the system, linked to the vehicle and (optionally) the rental contract

UC-03: Capture and Attach Photo Evidence

Field	Detail
Use Case ID	UC-03
Name	Capture and Attach Photo Evidence
Actors	Counter Agent
Preconditions	An inspection record is open (UC-02 completed); device camera is accessible
Main Success Scenario	1. System prompts agent to capture photos for each required angle (front, rear, left side, right side, interior) 2. Agent photographs each angle using the device camera 3. System automatically attaches a timestamp to each photo 4. System saves each photo against the open inspection record 5. Agent adds optional text notes per photo if damage is visible 6. Agent marks inspection as complete
Alternatives / Exceptions	2a. Camera permission denied — system prompts agent to enable camera access in device settings 4a. Upload fails due to poor connectivity — system queues photo locally and retries upload when connection is restored
Postconditions	Inspection record contains time-stamped photos for all required angles; record is marked complete and locked from further edits

UC-04: View Inspection Record / Dispute Evidence

Field	Detail
Use Case ID	UC-04

Name	View Inspection Record / Dispute Evidence
Actors	Counter Agent, Fleet Manager, Management/Owner
Preconditions	Actor is logged in; inspection records exist for the vehicle in question
Main Success Scenario	1. Actor searches for a vehicle by registration number or rental contract reference 2. System displays a list of past inspections for that vehicle 3. Actor selects an inspection record 4. System displays the inspection details: date/time, agent name, photos with timestamps, and any notes
Alternatives / Exceptions	1a. No records found — system displays 'No inspection records found' message
Postconditions	Actor has viewed the time-stamped photographic evidence for the selected inspection

UC-05: View Fleet Status Dashboard

Field	Detail
Use Case ID	UC-05
Name	View Fleet Status Dashboard
Actors	Fleet Manager, Management/Owner
Preconditions	Actor is logged in with Fleet Manager or Management role
Main Success Scenario	1. Actor navigates to the Fleet Dashboard 2. System displays a list of all vehicles with their current status (Available, Rented, Under Inspection) 3. Actor can filter by status or search by registration number 4. Actor can click a vehicle to view its inspection history summary
Alternatives / Exceptions	2a. No vehicles in system — system prompts admin to add vehicles
Postconditions	Actor has a current view of fleet status based on the most recent inspection and manual status data

UC-06: Manage Vehicles and Users (Admin)

Field	Detail
Use Case ID	UC-06
Name	Manage Vehicles and Users
Actors	Management/Owner
Preconditions	Actor is logged in with Management/Admin role

Main Success Scenario	1. Actor navigates to Administration panel 2. Actor can add, edit, or deactivate vehicles (registration number, make, model, year) 3. Actor can create, edit, or deactivate staff accounts (name, email, role)
Alternatives / Exceptions	3a. Attempting to deactivate a vehicle currently marked as Rented — system warns and requires confirmation
Postconditions	Vehicle and user records are updated in the system

C. User Stories

C1. Epics

- EP-01: Authentication and Access Control
- EP-02: Vehicle Inspection Workflow
- EP-03: Photo Evidence Capture and Storage
- EP-04: Inspection History and Dispute Evidence Retrieval
- EP-05: Fleet Status Dashboard
- EP-06: Vehicle and User Administration

C2. User Story List

All stories follow the format: As a <role>, I want <goal>, so that <benefit>. Acceptance criteria use Given/When/Then format.

ID	Role	Goal	Benefit	Acceptance Criteria	Priority
US-01	Counter Agent	log in with my email and password	I can access the system securely and only see features relevant to my role	Given valid credentials, when I submit, then I am taken to the inspection home screen. Given invalid credentials, when I submit, then I see a clear error and can retry.	Must

US-02	Counter Agent	start a new inspection for a specific vehicle	I can create a record linked to the right vehicle before capturing evidence	Given I am logged in, when I select 'New Inspection' and choose a vehicle by registration, then a new inspection record is created and I am taken to the photo screen.	Must
US-03	Counter Agent	capture photos of a vehicle at handover	there is time-stamped visual evidence of the vehicle's condition before the customer drives it	Given an open inspection, when I take a photo using the device camera, then the photo is saved with an automatic timestamp and linked to the inspection record.	Must
US-04	Counter Agent	capture photos of a vehicle at return	I can compare the vehicle's condition before and after the rental	Given a return inspection, when I capture photos, then they are saved and linked to the same rental contract reference as the handover inspection.	Must
US-05	Counter Agent	add a text note to a photo if I see damage	I can flag specific damage observations alongside the photographic evidence	Given a captured photo, when I type a note and save, then the note is stored alongside the photo in the inspection record.	Should

US-06	Counter Agent	mark an inspection as complete	the record is locked and cannot be altered after submission	Given all required photos are captured, when I tap 'Complete Inspection', then the record is locked, timestamped, and no further edits are possible.	Must
US-07	Counter Agent	search for a vehicle's inspection history by registration number	I can quickly find past records when a dispute arises at the desk	Given I search by a registration number, when the vehicle exists, then I see a list of past inspections ordered by date.	Must
US-08	Fleet Manager	view a dashboard of all vehicles and their current status	I can see at a glance which vehicles are available, rented, or under inspection	Given I am logged in as Fleet Manager, when I open the dashboard, then I see all vehicles with their current status updated from the latest inspection data.	Must
US-09	Fleet Manager	filter the fleet dashboard by vehicle status	I can quickly focus on vehicles that need attention	Given the dashboard is loaded, when I select a status filter (Available / Rented / Under Inspection), then only vehicles matching that status are shown.	Should

US-10	Fleet Manager	view the inspection history for a specific vehicle from the dashboard	I can investigate a vehicle's condition trend without leaving the fleet view	Given I click a vehicle on the dashboard, when the detail panel opens, then I see a list of past inspections with dates and completion status.	Should
US-11	Management/Owner	view time-stamped photo evidence for any inspection record	I can resolve damage disputes with factual, auditable evidence	Given I search for a vehicle, when I open an inspection record, then I see all photos with their capture timestamps and any agent notes.	Must
US-12	Management/Owner	add a new vehicle to the system	counter agents can inspect it immediately without waiting for a separate admin process	Given I am in the administration panel, when I enter a vehicle's registration, make, model and year and save, then the vehicle appears immediately in inspection and dashboard views.	Must
US-13	Management/Owner	create a staff account for a new counter agent or fleet manager	new staff can log in and use the system on their first day	Given I am in the administration panel, when I enter a name, email, and role and save, then the staff member receives login credentials and can access their role-appropriate features.	Must

US-14	Counter Agent	have my photo upload queued if I lose connectivity during an inspection	I don't lose evidence just because the internet dropped briefly	Given poor or no connectivity, when I capture a photo, then it is stored locally and automatically uploaded when connectivity is restored.	Could
US-15	Fleet Manager	manually update a vehicle's status if the system does not reflect the latest situation	the dashboard remains accurate even when an inspection wasn't completed	Given a vehicle status is incorrect, when I manually change it with a reason note, then the status updates and the change is logged with my name and timestamp.	Could

C3. Traceability — User Stories to Use Cases

Story ID	Story Summary	Linked Use Case(s)
US-01	Log in	UC-01
US-02	Start new inspection	UC-02
US-03	Capture handover photos	UC-03
US-04	Capture return photos	UC-03
US-05	Add damage note to photo	UC-03
US-06	Mark inspection complete	UC-03
US-07	Search inspection history	UC-04
US-08	View fleet dashboard	UC-05
US-09	Filter dashboard by status	UC-05

US-10	View vehicle inspection history from dashboard	UC-04, UC-05
US-11	View photo evidence for dispute	UC-04
US-12	Add new vehicle	UC-06
US-13	Create staff account	UC-06
US-14	Offline photo queue	UC-03
US-15	Manual status update	UC-05, UC-06

C4. Epics to Decompose Further in D3

- EP-02 (Inspection Workflow): US-02, US-03, US-04, US-05, US-06 need story-point estimation and may require further decomposition based on chosen tech stack.
- EP-03 (Photo Evidence): US-14 (offline queuing) may need to be split into separate 'capture locally' and 'sync on reconnect' stories.
- EP-05 (Fleet Dashboard): dashboard layout and filtering complexity should be clarified with client before Sprint 1.

D. Non-Functional Requirements

ID	Category	Requirement	Measure / Target
NFR-01	Usability	A counter agent should be able to complete a full inspection (all photos + notes + submit) with no prior training beyond a 10-minute walkthrough	Task completion in ≤ 5 minutes for a trained agent; validated via usability test in D3/D5
NFR-02	Usability	All primary inspection actions must be accessible on a standard smartphone screen without horizontal scrolling	No horizontal scroll on 375px-wide viewport (iPhone SE baseline)
NFR-03	Performance	Photo upload should complete within a reasonable time on typical mobile data speeds available in Accra	Upload of a 3MB photo completes within 15 seconds on a standard 4G connection
NFR-04	Performance	The fleet dashboard should load within 3 seconds for a fleet of up to 100 vehicles	Dashboard initial load ≤ 3 seconds measured with up to 100 vehicle records

NFR-05	Reliability	Captured inspection photos must not be permanently lost once captured by the agent	Photos are either uploaded immediately or queued locally with automatic retry; zero permanent data loss in normal operation
NFR-06	Security	Only authenticated users may access any part of the system	All routes/endpoints return 401 Unauthorized for unauthenticated requests
NFR-07	Security	A counter agent may not access fleet management or administration features	Role-based access control enforced server-side; UI does not expose inaccessible routes
NFR-08	Privacy	Vehicle inspection photos and rental contract references are treated as confidential business data	No inspection data exposed to unauthenticated requests; no real data committed to public repos or pasted into public AI tools
NFR-09	Maintainability	A developer unfamiliar with the codebase should be able to set up and run the project locally using the README and Makefile in under 30 minutes	Verified by at least one team member following setup instructions cold in D5
NFR-10	Accessibility	Core inspection workflow must be usable with standard mobile accessibility features enabled (e.g. large text, high contrast)	No critical UI elements broken when OS text size is set to 'Large'
NFR-11	Portability	The system must run in a standard modern mobile browser without requiring a native app install	Functional in Chrome (Android) and Safari (iOS) current versions
NFR-12	Data Retention	Inspection records must be retained for a minimum of 3 years to cover typical vehicle warranty and dispute periods	Records are not deleted automatically; deletion requires explicit admin action with confirmation

E. Data and External Interface Requirements

E1. Key Data Entities

Entity	Key Fields	Sensitivity
Vehicle	registration_number (PK), make, model, year, current_status (Available/Rented/Under Inspection), created_at, updated_at	Internal

User / Staff	user_id (PK), full_name, email, role (Counter Agent / Fleet Manager / Admin), is_active, created_at	Internal — email is PII, stored and accessed by admin only
RentalContract	contract_id (PK), vehicle_id (FK), customer_name, rental_start, rental_end, contract_reference	Confidential — customer name is PII
InspectionRecord	inspection_id (PK), vehicle_id (FK), contract_id (FK, optional), agent_id (FK), inspection_type (Handover/Return), status (Open/Complete), started_at, completed_at	Confidential
EvidencePhoto	photo_id (PK), inspection_id (FK), angle (Front/Rear/Left/Right/Interior), file_reference, timestamp, agent_notes	Confidential — photos may show vehicle damage and are tied to customer disputes
StatusChangeLog	log_id (PK), vehicle_id (FK), changed_by (FK), old_status, new_status, reason_note, changed_at	Internal

E2. External Systems, APIs and Services

- Device camera API — accessed via browser MediaDevices.getUserMedia() or equivalent; no external third party
- Cloud photo storage — exact provider TBD in D4 (e.g. Cloudinary free tier, AWS S3, Firebase Storage); photos stored as references in EvidencePhoto table
- Authentication provider — either custom JWT-based auth or a managed provider (e.g. Firebase Auth, Supabase Auth); decision in D4
- No payment system, SMS gateway, or email service is required this semester

E3. Data Sensitivity Classification

- Public: none (no public-facing data in this version)
- Internal: vehicle records, fleet status, inspection metadata
- Confidential: inspection photos, rental contract references, customer names
- Personal (PII): staff email addresses, customer names in rental contracts

E4. Data Retention and Deletion

- Inspection records and associated photos are retained for a minimum of 3 years (NFR-12)
- Deletion of any inspection record requires explicit admin action with a confirmation step and reason log
- Staff accounts are deactivated rather than deleted, to preserve attribution on past inspection records

- Customer PII in rental contract references is stored only as provided by the client's existing process; no additional PII is collected by InspectoFleet beyond what is necessary for dispute evidence

F. Glossary and Validation Evidence

F1. Glossary

Term	Definition
Counter Agent	A member of Kal Car Rentals staff who performs vehicle inspections at the handover (when a customer collects) and return (when a customer brings back) desk
Fleet Manager	A member of Kal Car Rentals staff responsible for monitoring the overall vehicle fleet status, location, and operational readiness
Handover Inspection	A vehicle inspection conducted when the customer collects the vehicle; documents the vehicle's condition before the rental begins
Return Inspection	A vehicle inspection conducted when the customer returns the vehicle; documents the vehicle's condition after the rental ends
Inspection Record	A digital record created in InspectoFleet for a single inspection event, containing photos, timestamps, agent notes, and status
Evidence Photo	A photograph taken during an inspection, stored with an automatic timestamp indicating when it was captured
Dispute	A disagreement between Kal Car Rentals and a customer about whether vehicle damage was pre-existing or caused by the customer during the rental period
Fleet Status	A per-vehicle indicator of its current operational state: Available, Rented, or Under Inspection
Rental Contract Reference	An identifier (e.g. contract number) linking an inspection record to the customer's rental agreement in Kal Car Rentals' existing records
Role-Based Access Control (RBAC)	A security model where each user account is assigned a role (Counter Agent, Fleet Manager, Admin) and can only access features permitted for that role
Time-Stamped	Automatically tagged with the exact date and time of capture by the system, not editable by the user
Turnaround	The process of receiving a returned vehicle, inspecting it, and preparing it for the next rental

F2. Client Validation Notes

Requirements baseline validated with: Phaniel K. A. Wagba (Director, Kal Car Rentals) via initial client conversation (evidence on file, see docs/01_proposal/).

- Confirmed: paper inspection process, photo evidence gap, and damage dispute problem are accurately described
- Confirmed: counter agents use mobile phones during turnarounds; mobile-first design is appropriate
- Confirmed: fleet visibility is a secondary but real concern
- To be confirmed before D3: exact number of vehicles in the fleet; number of counter agents; whether customers are ever given access to their own inspection records; whether the client uses any existing software for contract management that InspectoFleet should reference

F3. Open Questions (to resolve before D3/D4)

- OQ-01: How many vehicles does Kal Car Rentals currently operate? (Affects dashboard performance targets and database sizing)
- OQ-02: How many counter agents will use the system simultaneously? (Affects concurrent user load assumptions)
- OQ-03: Does the client have an existing contract management system whose IDs should be used as rental contract references, or is this a free-text field?
- OQ-04: Should customers be able to view their own inspection records (e.g. via a link)? (Scope decision — currently scoped out but worth confirming)
- OQ-05: What cloud provider/hosting is acceptable to the client for photo storage? Are there data residency preferences?